Combinatorics

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1 Combinatorics (組合數學)

I Counting principle (計數原理) or fundamental principle of counting (基本計數原理)

i Rule of sum or addition principle (加法原理)

If there are a ways to do something and b ways to do another thing, and the two things can't both be done, then there are $a + b$ total possible ways to do one of the things.

ii Rule of product or multiplication principle (乘法原理)

If there are a possible ways to do something and b possible ways to do another thing, then there are $a \cdot b$ total possible ways to do both things.

iii Complement laws

$$(A')' = A.$$

$$A \setminus B = A \cap B'.$$

$$A \subseteq B \iff B' \subseteq A'.$$

iv De Morgan's laws

$$(A \cup B)' = A' \cap B'.$$

$$(A \cap B)' = A' \cup B'.$$

v Counting principle for power set (冪集計數原理)

$$\left|2^A\right| = 2^{|A|}$$

vi Principle of inclusion-exclusion (PIE) or inclusion–exclusion principle (排容原理/取捨原理)

For indexed family $(A_i)_{i=1}^n$ with $n \in \mathbb{N}$,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{J \subseteq \mathbb{N}_{\leq n} \wedge J \neq \emptyset} (-1)^{|J|+1} \left|\bigcap_{j \in J} A_j\right|.$$

Proof. When $n = 1$,

$$|A_1| = |A_1|.$$

When $n = 2$,

$$\left|A_1 \cup A_2\right| = \left|A_1 \setminus (A_1 \cap A_2)\right| + \left|A_2 \setminus (A_1 \cap A_2)\right| + \left|A_1 \cap A_2\right| = \left|A_1\right| + \left|A_2\right| - \left|A_1 \cap A_2\right|.\}$$

Assume that is holds for $n \in \mathbb{N}_{\geq 2}$. For $n+1$,

$$\begin{aligned}\left|\bigcup_{i=1}^{n+1} A_i\right| &= \left|\bigcup_{i=1}^{n} A_i\right| + \left|A_{n+1}\right| - \left|\left(\bigcup_{i=1}^{n} A_i\right) \cap A_{n+1}\right| \\ &= \sum_{i=1}^{n} \left|A_i\right| - \sum_{1\leq i<j\leq n} \left|A_i \cap A_j\right| + \sum_{1\leq i<j<k\leq n} \left|A_i \cap A_j \cap A_k\right| - \cdots + (-1)^{n+1} \left|\bigcap_{i=1}^{n} A_i\right| + \left|A_{n+1}\right| - \left(\sum_{i=1}^{n} \left|A_i \cap \right.\right. \\ &= \sum_{i=1}^{n+1} \left|A_i\right| - \sum_{1\leq i<j\leq n+1} \left|A_i \cap A_j\right| + \sum_{1\leq i<j<k\leq n+1} \left|A_i \cap A_j \cap A_k\right| - \cdots + (-1)^{n+2} \left|\bigcap_{i=1}^{n+1} A_i\right|\end{aligned}$$

By mathematical induction, the statement is true for all $n \in \mathbb{N}$. □

vii Pigeonhole principle (鴿籠原理/鴿洞原理/鴿巢原理) or Dirichlet's drawer principle (狄利克雷抽屜原理)

Given $n, k \in \mathbb{N}$, if k pigeons fly into n cages, then there must be at least one cage containing at least $\lceil \frac{k}{n} \rceil$ pigeons.

viii Generalized pigeonhole principle

Given $n, k_1, k_2, \ldots, k_n \in \mathbb{N}$, if $\sum_{i=1}^{n} k_i - n + 1$ pigeons fly into n cages, then there exists at least one $i \in \mathbb{N}_{\leq n}$ such that the ith cage contains at least k_i pigeons.

II Factorial (階乘)

i Factorial

$$n! \coloneqq \prod_{i=1}^{n} i, \quad n \in \mathbb{N}_0.$$

ii Falling factorial

$$(n)_m \coloneqq \prod_{i=0}^{m-1} (n-i), \quad n \in \mathbb{C} \wedge m \in \mathbb{N}.$$

III Gamma function (伽瑪函數)

i Definition

$$\Gamma(z) \coloneqq \int_0^\infty t^{z-1} e^{-t} \, \mathrm{d}t, \quad \Re(z) > -1.$$

The integral diverges if $\Re(z) \leq -1$.

Proof. TODO □

ii Recurrence

$$\Gamma(z+1) = z\Gamma(z), \quad \Re(z) > -1.$$

Proof. TODO □

For $z \in \mathbb{N}$,

$$\Gamma(z) = (z-1)!.$$

For $m \in \mathbb{N} \land z \notin \mathbb{N}_{\leq m}$,

$$\frac{\Gamma(z)}{\Gamma(z-m)} = (z-1)_m.$$

iii TODO

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\mathsf{csc}(\pi z)}$$

Proof. TODO □

iv TODO

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

Proof. TODO □

IV 直線排列

i 盡相異物排列

從 n 相異物中取 m 個排成一直列（$0 \leq m \leq n$），正逆序視為二，其排列總數 $P_m^n = \frac{n!}{(n-m)!}$。

ii 不盡相異物排列

設有 n 物，共有 k 種，第 i 種有 m_i 個。全取排成一直列，正逆序視為二，其排列總數為 $\frac{n!}{\prod_{i=1}^{n} m_i!}$。

iii 盡相異物重複排列

從 n 類相異物中，任取 m 個排成一直列，正逆序視為二，其中每類物品的個數均不小於 m 且可重複選取，其排列總數為 n^m。

iv 錯排（Derangement）

n 相異物全取作直線排列，其中 m 物（$m \leq n$）依次被限制不能排列於相異單一指定位置之排列，其方法數稱 D_m^n；當 $m = n$，錯排方法數稱 D_n。

v　全錯排

遞迴式：

$$\begin{cases} D_0 &= 1 \\ D_1 &= 0 \\ D_n &= (n-1)\cdot(D_{n-1}+D_{n-2}) \quad (\text{當 } n>2) \end{cases}$$

一般式：

$$\begin{aligned} D_n &= n!\cdot\left(1+\sum_{i=1}^{n}\frac{(-1)^i}{i!}\right) \\ &= \sum_{i=0}^{n}(C_i^n\cdot(n-i)!\cdot(-1)^i \end{aligned}$$

錯排機率：

$$\lim_{n\to\infty}\frac{D_n}{n!}=\frac{1}{e}$$

vi　非全錯排

$$D_m^n=\sum_{i=0}^{m}\left((-1)^i\cdot C_i^m\cdot(n-i)!\right)$$

V　環狀排列（Cyclic permutation）

環狀排列：n 物全取排列成環狀，不同方法之判定僅考慮相對位置，不考慮絕對位置，惟翻轉（順時針 $\rightleftharpoons$ 逆時針）視為二種。

i　盡相異物環狀排列

$$\frac{n!}{m\cdot(n-m)!}$$

ii　不盡相異物環狀排列

n 不盡相異物，全取作環狀排列，求其方法數。

- **法一：**
  子循環：一個長度 n 的給定直線排列，若其前 $\frac{n}{m}$ 物重複 m 次等同於原排列，且不存在 >1 的 k 使原排列的前 $\frac{n}{m\cdot k}$ 物重複 k 次等同於原排列的前 $\frac{n}{m}$ 物，則稱原排列的前 $\frac{n}{m}$ 物為一個長度 $\frac{n}{m}$ 的子循環，稱原排列之子循環長度為 $\frac{n}{m}$、子循環數目為 m。

  令該 n 不盡相異物的所有可能直線排列中，有 d_i 個之子循環長度為 ℓ_i，且 $\sum_{i=1}^{m}d_i=$ 該 n 不盡相異物直線排列方法數，則：

$$\text{所求}=\sum_{i=1}^{m}\frac{d_i}{\ell_i}$$

- **法二：**
  令該 n 不盡相異物可分為 k 相異類，第 i 類（$1 \le i \le k$）有 m_i 件相同物。最大公因數 $\gcd(m_1, m_2, m_3, \dots, m_k) = g$。令所求為 R。
  - 最大公因數 g 為 1：
    $$R = \frac{(n-1)!}{\prod_{i=1}^{k}(m_i!)}$$
  - 最大公因數 g 為一質數 p：
    $$R = \frac{1}{n} \cdot \left(\frac{n!}{\prod_{i=1}^{k}(m_i!)} - \frac{n!}{\prod_{i=1}^{k}\left(\frac{m_i}{p}!\right)} \right) + \frac{\left(\frac{n}{p}-1\right)!}{\prod_{i=1}^{k}\left(\frac{m_i}{p}!\right)}$$
  - 最大公因數 g 為任一正整數：
    令 g之所有相異質因數由小到大依序為$p_1, p_2, p_3, \dots, p_q$，
    $$R = \sum_{j=0}^{q} \left((-1)^j \cdot \sum_{|Y|=j} \frac{\left(\sum_{i=1}^{k} \frac{m_i}{\prod_{y \in Y} y}\right)!}{\prod_{i=1}^{k}\left(\frac{m_i}{\prod_{y \in Y} y}!\right)} \right)$$
    ，其中$Y \subseteq \{p_1, p_2, p_3, \dots, p_q\}$
    ，定義 $\prod_{y \in \emptyset} y = 1$

VI Choose with replacement

Choose with replacement refers to a selection process where each item can be selected multiple times, because after you pick an item, you put it back before the next pick.

i Distinguishable objects choose with replacement

The number of ways to choose with replacement m objects out of n distinguishable objects is

$$m^n.$$

VII Binomial coefficients (二項次係數) or combinations (組合數)

Below, we will define binomial coefficient $\binom{n}{m}$, also known as combination C_m^n, called n choose m.

i Nonnegative integer

The number of ways to choose m objects out of n distinguishable objects is

$$\frac{n!}{m!(n-m)!}.$$

For all $n, m \in \mathbb{N}_0$, define

$$\binom{n}{m} = \begin{cases} \dfrac{n!}{m!(n-m)!}, & m \leq n \\ 0, & \text{otherwise} \end{cases}.$$

ii Complex number n and nonnegative integer m

For all $n \in \mathbb{C}$ and $m \in \mathbb{N}_0$, define

$$\binom{n}{m} = \frac{\prod_{i=0}^{m-1}(n-i)}{m!}.$$

It satisfies

$$\binom{n}{m} = (-1)^m \cdot \binom{-n+m-1}{m}.$$

iii Real part of m and real part of $n-m$ greater than -1

For all $n, m \in \mathbb{C}$ with $\Re(m), \Re(n-m) > -1$,

$$\binom{n}{m} = \frac{\Gamma(n+1)}{\Gamma(m+1)\Gamma(n-m+1)}.$$

iv Complex number

For all $n, m \in \mathbb{C}$,

$$\binom{n}{m} = \begin{cases} \dfrac{\Gamma(n+1)}{\Gamma(m+1)\Gamma(n-m+1)}, & \Re(m), \Re(n-m) > -1 \\ (-1)^m \cdot \binom{-n+m-1}{m}, & \Re(n) \leq -1 \\ 0, & \text{otherwise} \end{cases}.$$

v Binomial theorem (二項式定理)

For $n, x \in \mathbb{C}$, if either

- $|x| < 1$, or
- $\Re(n) > -1$ and $x = 1$, or
- $m \geq 0$ and $|x| = 1$, or
- $m \in \mathbb{N}_0$,

then:

$$(1+x)^n = \sum_{m=0}^{\infty} \binom{n}{m} x^m,$$

where the RHS is called binomial series.

Proof. TODO □

VIII Multinomial coefficient

i Nonnegative integer

The number of ways to partition n distinguishable objects into m labeled groups with size $n_1, n_2, \dots, n_m$ is

$$\frac{n!}{\prod_{i=1}^{m} n_i!}.$$

For all $n, m, n_1, n_2, \dots, n_m \in \mathbb{N}_0$ with $\sum_{i=1}^{m} n_i = n$, define

$$\binom{n}{n_1, n_2, \dots, n_m} = \begin{cases} \frac{n!}{\prod_{i=1}^{m} n_i!}, & m \leq n \\ 0, & \text{otherwise} \end{cases}.$$

The number of ways to partition k of n distinguishable objects into m labeled groups with size $n_1, n_2, \dots, n_m$ is

$$\frac{(n)_k}{\prod_{i=1}^{m} n_i!}.$$

For all $n, m, n_1, n_2, \dots, n_m \in \mathbb{N}_0$, let $k = \sum_{i=1}^{m} n_i \leq n$, define

$$\binom{n}{n_1, n_2, \dots, n_m} = \begin{cases} \frac{(n)_k}{\prod_{i=1}^{m} n_i!}, & m \leq n \\ 0, & \text{otherwise} \end{cases}.$$

ii Complex number n and nonnegative integer n_i

For all $n \in \mathbb{C}$ and $m, n_1, n_2, \dots, n_m \in \mathbb{N}_0$, let $k = \sum_{i=1}^{m} n_i$, define

$$\binom{n}{n_1, n_2, \dots, n_m} = \frac{(n)_k}{\prod_{i=1}^{m} n_i!}.$$

It satisfies

$$\binom{n}{n_1, n_2, \dots, n_m} = (-1)^k \cdot \binom{-n+m-1}{n_1, n_2, \dots, n_m}.$$

iii Real part of n_i and real part of $n - k$ greater than -1

For all $m \in \mathbb{N}_0$ and $n, n_1, n_2, \dots, n_m \in \mathbb{C}$ with $\Re(n_i), \Re(n-k) > -1$, let $k = \sum_{i=1}^{m} n_i$, define

$$\binom{n}{n_1, n_2, \dots, n_m} = \frac{\Gamma(n+1)}{\prod_{i=1}^{m} \Gamma(n_i+1)\Gamma(n-k+1)}.$$

iv Complex number

For all $m \in \mathbb{N}_0$ and $n, n_1, n_2, \ldots, n_m \in \mathbb{C}$, let $k = \sum_{i=1}^{m} n_i$, define

$$\binom{n}{n_1, n_2, \ldots, n_m} = \begin{cases} \dfrac{\Gamma(n+1)}{\prod_{i=1}^{m} \Gamma(n_i+1)\Gamma(n-k+1)}, & \Re(n_i), \Re(n-k) > -1 \\ (-1)^k \cdot \dbinom{-n+m-1}{n_1, n_2, \ldots, n_m}, & \Re(n_i), \Re(-n+m-1-k) > -1 \cdot \\ 0, & \text{otherwise} \end{cases}$$

v Multinomial theorem (多項式定理)

For $n, x_1, x_2, \ldots, x_m \in \mathbb{C}$, if $\sum_{i=1}^{m} |x_i| < 1$, then:

$$\left(\sum_{i=1}^{m} x_i \right)^n = \sum_{n_1, n_2, \ldots, n_m \geq 0} \binom{n}{n_1, n_2, \ldots, n_m} \prod_{i=1}^{m} {x_i}^{n_i},$$

where the RHS is called multinomial series.

Proof. TODO □

IX 重複組合

由 n 類相異物中，任取 m 個為一組之方法數，其中每類物品的個數均不小於 m 且可重複選取。記作 H_m^n。

$$\begin{aligned} H_m^n &= C_m^{n+m-1} \\ &= C_{n-1}^{n+m-1} \end{aligned}$$

X 組合恆等式

i 范德蒙恆等式（Vandermonde identity）

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

Proof.
Consider a lattice path from $(0,0)$ to (n,n) in a grid where each step moves either right $(1,0)$ or up $(0,1)$.
Since we must choose n steps to move right out of the total $2n$ steps, the total number of such paths is given by:

$$\binom{2n}{n}.$$

Now, let’s count the same paths differently. Choose an intermediate point $(k, n-k)$ at step n. The number of ways to reach this point from $(0,0)$ using k right steps and $n-k$ up steps is $\binom{n}{k}$. From $(k, n-k)$ to (n,n), we need $n-k$ right steps and k up steps, which can be done in $\binom{n}{k}$ ways. Summing over all possible k, we obtain:

$$\sum_{k=0}^{n} \binom{n}{k}^2.$$

□

ii 巴斯卡公式（Pascal’s formula）

$$C_m^n = C_{m-1}^{n-1} + C_m^{n-1}$$

iii 組合恆等式

$$\sum_{k=0}^{n} C_k^n = 2^n$$

$$\sum_{k=0}^{n} (-1)^k \cdot C_k^n = 0$$

$$\sum_{k=0}^{n} 2^k \cdot C_k^n = 3^n$$

$$\sum_{k=0}^{n} k \cdot C_k^n = n \cdot 2^{n-1}$$

$$\sum_{k=0}^{n} k^2 \cdot C_k^n = n \cdot (n-1) \cdot 2^{n-2} + n \cdot 2^{n-1}$$

$$\sum_{k=3}^{n} C_2^k = n \cdot 2^{n-1}$$

$$\left(\prod_{i=0}^{x} (k-i)\right) \cdot C_k^n = \left(\prod_{i=0}^{x} (n-i)\right) \cdot C_{k-x-1}^{n-k-1} \quad x < k \wedge x \in \mathbb{N}$$

$$\sum_{k=0}^{n} \left(\left(\prod_{i=0}^{x} (k-i)\right) \cdot C_k^n\right) = \sum_{k=0}^{n} \left(\left(\prod_{i=0}^{x} (n-i)\right) \cdot C_{k-x-1}^{n-k-1}\right)$$

$$= \frac{n!}{(n-x-1)!} \cdot 2^{n-x-1}$$

$$x < n \wedge x \in \mathbb{N}, \text{定義} \forall k < 0 : C_k^n = 0$$

iv 巴斯卡三角形（Pascal's triangle）/楊輝三角形

令最上面一列為第 0 列，向下每列遞增 1；每一列最左之數為第 0 個，向右每個遞增 1。巴斯卡三角形之第 n 列第 m 個數定義為 C_m^n。

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \\ \vdots & & & & & \vdots & \end{array}$$

XI 卡特蘭數（Catalan number）

i 卡特蘭數（Catalan number）

所有在 $n \times n$ 格點中不越過對角線的單調路徑的個數，一個單調路徑從格點左下角出發，在格點右上角結束，每一步均為向上或向右，記作 C_n。

遞迴式：

$$\begin{cases} C_0 &= 1 \\ C_n &= \sum_{k=1}^{n} C_{k-1} \cdot C_{n-k} \quad (\text{當 } n > 0) \end{cases}$$

Proof. 令對角線為 $y = x$，第一次到達對角線上時的位置為 (k, k)，則第一次到達對角線前的單調路徑數為 C_k，第一次到達對角線後的單調路徑數為 C_{n-k}，故對 $n > 0$ 成立。$C_0 = C_1 = 0$，亦成立。 □

一般式：

$$\begin{aligned} C_n &= C_n^{2n} - C_{n-1}^{2n} \\ &= \frac{1}{n+1} \cdot C_n^{2n} \end{aligned}$$

Proof. 自 $(0, 0)$ 至 (n, n) 的單調路徑數為 C_n^{2n}。考慮這些路徑中不符合卡特蘭數定義者，其第一次跨越對角線 $y = x$ 的點必在 $y = x + 1$ 上，將接下來的路徑對 $y = x + 1$ 鏡射，則終點 (n, n) 變為 $(n - 1, n + 1)$。在此 $(n - 1) \times (n + 1)$ 格點中自 $(0, 0)$ 至 $(n - 1, n + 1)$ 的單調路徑數為 C_{n-1}^{2n}。故 $C_n = C_n^{2n} - C_{n-1}^{2n} = \frac{1}{n+1} \cdot C_n^{2n}$。 □

ii 非正方形格點的卡特蘭數

所有在 $n \times k$ 格點中不越過對角線的單調路徑的個數，一個單調路徑從格點左下角出發，在格點右上角結束，每一步均為向上或向右，記作 $C_{n,k}$。

一般式：

$$C_{n,k} = \begin{cases} C_k^{n+k} - C_{k-1}^{n+k} & (\text{當 } n \geq k) \\ C_n^{n+k} - C_{n-1}^{n+k} & (\text{當 } n \leq k) \end{cases}$$

XII Stirling number of the first kind

i Definition

The (signed) Stirling number of the first kind, denoted as $s(n, k)$, is defined for all $n, k \in \mathbb{N}_0$ such that

$$\prod_{j=0}^{n-1}(x - j) = \sum_{k=0}^{\infty} s(n, k)x^k$$

The unsigned Stirling number of the first kind, denoted as $\begin{bmatrix} n \\ k \end{bmatrix}$, is the absolute value of the signed one.

ii Signs

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}$$

Proof. PLACEHOLDER □

iii Rising factorial

For all $n, k \in \mathbb{N}_0 \wedge n \geq k$

$$\prod_{j=0}^{n-1}(x + j) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} x^k$$

Proof. PLACEHOLDER □

iv Recurrence relation

$$s(n + 1, k) = -ns(n, k) + s(n, k - 1)$$

with base case

$$s(0, 0) = 1$$

$$s(n, 0) = s(0, n) = 0, \quad n \in \mathbb{N}$$

Proof. PLACEHOLDER □

$$\begin{bmatrix} n + 1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k - 1 \end{bmatrix}$$

with base case

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$$

$$\begin{bmatrix} n \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ n \end{bmatrix} = 0, \quad n \in \mathbb{N}$$

Proof. PLACEHOLDER □

v Properties

$$s(n,k) = 0, \quad n < k$$

$$s(n,1) = (-1)^{n-1}(n-1)!, \quad n \in \mathbb{N}$$

$$s(n,n) = 1$$

$$s(n,n-1) = -\sum_{j=1}^{n-1} j = -\frac{n(n-1)}{2} = -\binom{n}{2}, \quad n \in \mathbb{N}$$

XIII Stirling number of the second kind

i Definition

The Stirling number of the second kind, denoted as $S(n,k)$ or $\left\{ {n \atop k} \right\}$, counts the number of different equivalence relations with exactly k equivalence classes that can be defined on an n element set, for all $n, k \in \mathbb{N}_0$.

ii Recurrence relation

$$S(n+1,k) = kS(n,k) + S(n,k-1)$$

with base case

$$S(0,0) = 1$$

$$S(n,0) = S(0,n) = 0, \quad n \in \mathbb{N}$$

Proof. PLACEHOLDER □

iii Properties

$$S(n,k) = 0, \quad n < k$$

$$S(n,1) = 1, \quad n \in \mathbb{N}$$

$$S(n,n) = 1$$

$$S(n,n-1) = \sum_{j=1}^{n-1} j = \frac{n(n-1)}{2} = \binom{n}{2}, \quad n \in \mathbb{N}$$